

① The ratio of the magnitudes of the electric field and magnetic field of a plane electromagnetic wave is

- (A) 1 (B) $\frac{1}{c}$ (C) $\frac{1}{c^2}$ (D) c

② Specify the transition of electron in the wavelength of the line in the Bohr model of hydrogen atom which gives rise to the spectral line of highest wavelength.

- (A) $n=3$ to $n=1$ (B) $n=3$ to $n=2$
(C) $n=4$ to $n=1$ (D) $n=4$ to $n=2$

③ A concave mirror of focal length 12cm. forms a three times magnified image virtual one of an object. Find the distance of the object from the mirror.

④ which one of the following elements will require the highest energy to take out an electron from them?

Pb, Ge, C and Si

- (A) Pb (B) Ge (C) C (D) Si

⑤ A steady current of 8mA flows through a wire. The number of electrons passing through the cross section of the wire in 10 seconds is?

⑥ An isolated point charge produces an electric field $\vec{E}$ at a point 3m away from it. The distance of the point at which the field will be $\frac{\vec{E}}{4}$ is?

- (A) 2m (B) 3m (C) 4m (D) 6m

⑦ The current in the primary coil of a pair of coils changes from 7A to 3A in 0.4s. The mutual inductance between the two coils is 0.5H. The induced emf in the secondary coil is:-

- (A) 50V (B) 75V (C) 100V (D) 220V

⑧ The electric potential V at any point (x, y, z) is given by $V = 3x^2$, where x is in metres, and V is in volts. The electric field at the point $(1m, 0, 2m)$ is:

(A) $6 \frac{V}{m}$ along $-x$ axis.

(B) $6 \frac{V}{m}$ along $+x$ -axis

(C) $1.5 \frac{V}{m}$ along $-x$ axis

(D) $1.5 \frac{V}{m}$ along $+x$ axis.